

Question:

An 8-year-old girl is brought to the emergency department because of a 2-day history of low-grade fever, itchy rash, and generalized joint pain. The rash initially started in the antecubital and popliteal fossae and then spread to her trunk and distal extremities. One week ago, she was diagnosed with acute sinusitis and was started on amoxicillin. She has no history of adverse drug reactions and immunizations are up-to-date. Her temperature is 37.5°C (99.5°F), pulse is 90/min, and blood pressure is 110/70 mm Hg. Physical examination shows periorbital edema and multiple erythematous, annular plaques of variable sizes over her entire body. One of the lesions in the right popliteal fossa has an area of central clearing and the patient's mother reports that it has been present for over 24 hours. Urinalysis is normal. Which of the following is the most likely diagnosis?

Answer Choices:

- A. Serum sickness-like reaction
- B. Stevens-Johnson syndrome
- C. Pemphigus vulgaris
- D. Drug reaction with eosinophilia and systemic symptoms

Reasoning + Answer:

<think> The patient is presenting with a rash that started in the antecubital and popliteal fossae and then spread to her trunk and distal extremities, which is consistent with a symmetrical distribution.

The patient was recently started on amoxicillin, which suggests a possible drug-related reaction.

The physical examination shows periorbital edema and multiple erythematous, annular plaques, which is consistent with a serum sickness-like reaction.

The patient's urinalysis is normal, which makes Stevens-Johnson syndrome less likely, as it is typically associated with renal involvement.

Pemphigus vulgaris is an autoimmune disease that affects the skin and mucous membranes, but it is not consistent with the patient's symptoms and history.

Drug reaction with eosinophilia and systemic symptoms is a possible diagnosis, but it is not the most likely diagnosis given the patient's symptoms and physical examination findings. </think><answer>Serum sickness-like reaction</answer>